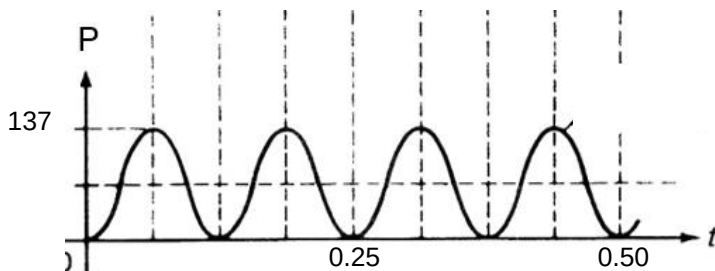


5(ii)	1. no effect. No effect. 2. no effect, radius is halved	B1 B1
6 (a)	Root-mean-square value of an alternating current is defined as the value of an alternating current which would dissipate heat at the <u>same average heating effect /rate in a given resistance as a steady direct current</u> of that value.	B1
(b)(i)	For e.m.f. to be maximum, $\sin(\omega t) = 1$. Hence, max e.m.f. $E_o = NBA(2\pi f)$ $= (800)(0.50)(5.0 \times 10^{-2} \times 8.0 \times 10^{-2})(2\pi)(240/60)$ $= 40.2 \text{ V}$	C1 A1
(ii)	$I_o = \frac{V_o}{R} = \frac{40.2}{0.60 + 11} = 3.4655 \approx 3.47 \text{ A}$ $I_{rms} = \frac{I_o}{\sqrt{2}} = 2.45 \text{ A}$	C1 A1
(iii)	 Correct shape with 2 cycles drawn. Label $P_{\max} = I_o^2 R = (3.4655^2)(11.4) = \underline{137} \text{ W}$ Label period $T = \frac{60}{240} = 0.25 \text{ s}$	B1 B1 B1

7 (a)	vibration / oscillation parallel to energy transmission / wave	D1
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